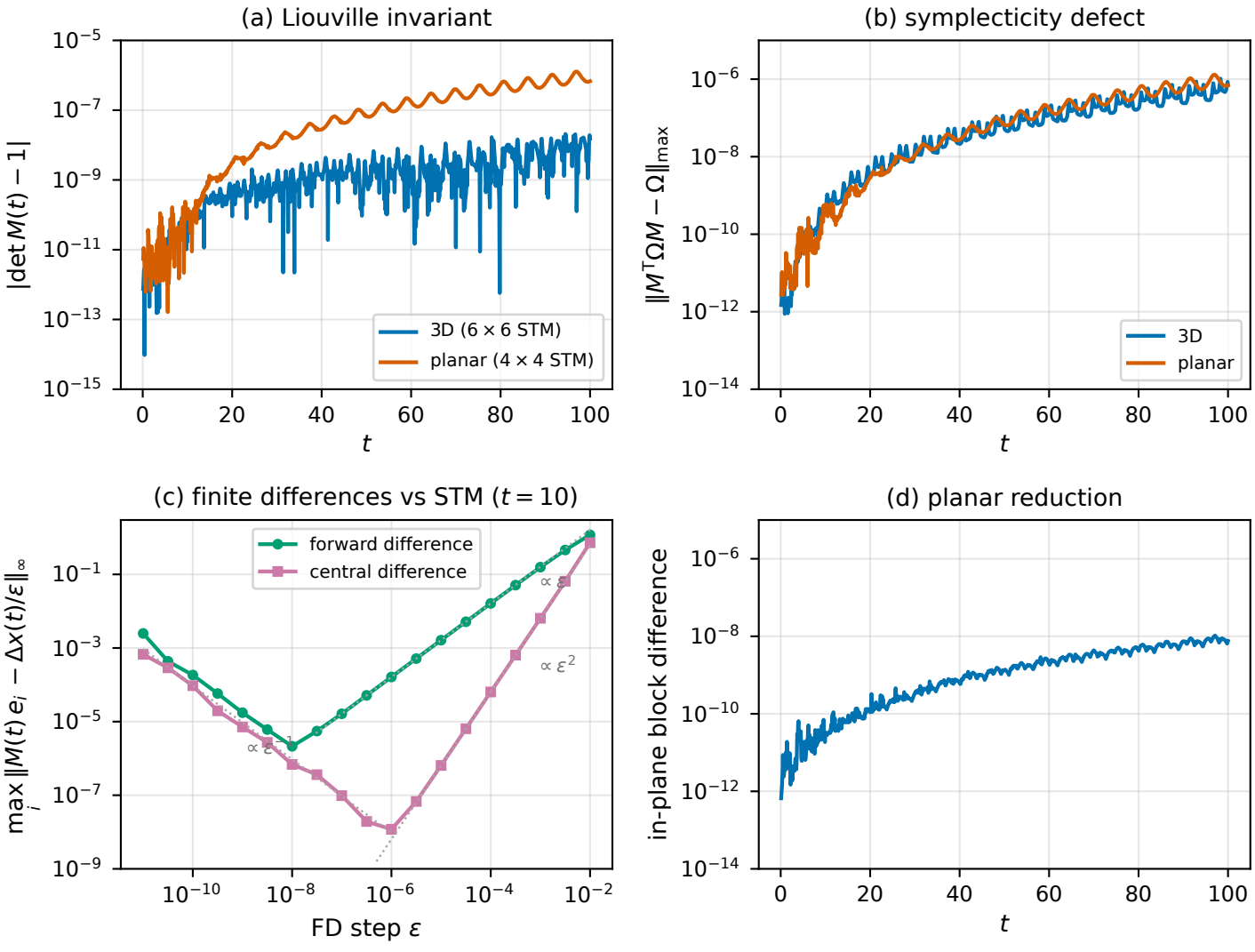
(a) Liouville invariant
$|\det M(t) - 1|$
t
3D (6 × 6 STM)
planar (4 × 4 STM)
(b) symplecticity defect
$\|M^{\mathsf{T}}\Omega M - \Omega\|_{\max}$
t
3D
planar
(c) finite differences vs STM ($t = 10$)
$\max_i \|M(t)\, e_i - \Delta x(t)/\varepsilon\|_\infty$
forward difference
central difference
$\propto \varepsilon^{-1}$
$\propto \varepsilon$
$\propto \varepsilon^2$
FD step ε
(d) planar reduction
in-plane block difference
t